

Exact diagonalization of the quantum spin ice gauge sector beyond quantum Monte Carlo: removing torus-winding contamination by zero-transport projection

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Abstract

The thermodynamics of quantum spin ice (QSI) is governed by an emergent compact U(1) gauge field. Its unfrustrated 0-*flux* sector ($J_{\pm} > 0$) is accessible to sign-problem-free quantum Monte Carlo (QMC), but the frustrated π -*flux* sector ($J_{\pm} < 0$) realized in the dipolar-octupolar cerium pyrochlores carries a QMC sign problem and has no controlled finite-temperature benchmark. Small-cluster exact diagonalization (ED) is the natural alternative – but on the smallest periodic clusters it is contaminated by ice-preserving four-site cycles that close only by winding the cluster torus. They enter the effective Hamiltonian at *second* order, $g_4 = 4J_{\pm}^2/J_{zz}$, parametrically above the genuine photon coupling $g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2$. It is tempting to remove them by averaging the ED over U(1) boundary twists $\boldsymbol{\varphi}$. We show that this fails for two independent reasons, both confirmed by exact numerics, and we construct the projector that succeeds. (i) *Twist phases attach to virtual paths, not to loops.* A ring exchange over a cycle $\mathcal{C}$ is generated by the *two perfect matchings* of $\mathcal{C}$ into nearest-neighbour dimer moves; its coupling at twist $\boldsymbol{\varphi}$ is $c_{\mathcal{C}}(\boldsymbol{\varphi}) = -\frac{g_4}{2} [e^{-i\mathbf{N}_A \cdot \boldsymbol{\varphi}} + e^{-i\mathbf{N}_B \cdot \boldsymbol{\varphi}}]$ with $\mathbf{N}_A + \mathbf{N}_B = \mathbf{w}$ the loop winding. On both production clusters [$1 \times 1 \times 1$ cubic (16 sites) and $2 \times 2 \times 2$ FCC (32 sites)] *every* winding four-loop and every wrapping hexagon has one matching carrying zero net wrap ($\mathbf{N}_A = \mathbf{0}$, generically two bulk-bond dimer moves), which no boundary twist can dephase: even a perfect operator-level twist average retains exactly *half* of every spurious coupling. A gauge-invariant corollary, $|c_{\mathcal{C}}| \propto |1 + e^{-i\mathbf{w} \cdot \boldsymbol{\varphi}}|$, means individual couplings switch *off* (not sign-flip) at twist corners with $\mathbf{w} \cdot \mathbf{p}$ odd – verified level-by-level against exact corner spectra. (ii) *Averaging observables is weaker still:* the free energy is nonlinear in H_{eff} , so corner-averaging $C(T, \boldsymbol{\varphi})$ cancels only terms odd in the spurious couplings; even powers survive at full strength. Exactly on the 16-site cluster at $J_{\pm} = -0.1J_{zz}$, the low- T peak moves only from $0.037J_{zz}$ (bare) to $0.031J_{zz}$ (corner-averaged observable) and to $0.018J_{zz}$ (corner-averaged *operator*, obtained by exact Löwdin downfolding at each corner – the half-coupling scale $g_4/2 = 0.02J_{zz}$), never approaching the photon scale. The projector that works keeps only processes with zero *net charge transport* $\boldsymbol{\delta} = \mathbf{0}$ – the gauge-invariant label that a uniform emergent vector potential couples to. It kills every winding four-loop and wrapping hexagon exactly, at all orders, while preserving every contractible hexagon at full strength (verified to machine precision on both clusters), and it moves the 16-site peak to $0.0075J_{zz}$, at the hexagon scale $g_{\text{hex}} = 0.012J_{zz}$, an order of magnitude below the bare artifact. We give exact corrected corner spectra (the ground state at $\boldsymbol{\varphi} = \mathbf{0}$ is a singlet; previously quoted “7-fold ground manifolds” were Lanczos artifacts), the combinatorial censuses, and practical recipes: path-selected effective Hamiltonians in perturbation theory, and a nonperturbative dipole mask on the exactly downfolded Hamiltonian beyond it. Because the same selection rule cleans not only the Hamiltonian but every ice-manifold operator, the projected cluster becomes a quantitative laboratory for the emergent gauge sector: we outline how it delivers benchmarked π -flux gauge thermodynamics – validated against 0-flux QMC and extended to the sign-problem regime QMC cannot reach – with the flux-locked thermodynamic response of the emergent photon as the target observable.

1 The finite-size short-loop problem

The XXZ pyrochlore Hamiltonian in local axes is

$$H_{\text{XXZ}} = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad J_{\pm} = -\frac{1}{4}(J_{xx} + J_{yy}), \quad (1)$$

where $\langle ij \rangle$ runs over nearest-neighbour (NN) bonds. For $|J_{\pm}|/J_{zz} \ll 1$, third-order perturbation theory on the 2-in-2-out (ice-rule) manifold reduces eq. (1) to the Hermele–Fisher–Balents ring-exchange Hamiltonian [1, 2]

$$H_{\text{eff}}^{(\text{HFB})} = -\frac{12 J_{\pm}^3}{J_{zz}^2} \sum_{\text{hex}} (S^+ S^- S^+ S^- S^+ S^-)_{\text{hex}} + \text{h.c.}, \quad (2)$$

summed over hexagonal plaquettes; $g_{\text{hex}} \equiv 12|J_{\pm}|^3/J_{zz}^2$ sets the emergent-photon scale. The sign of the coefficient is locked to $\text{sgn}(J_{\pm}^3)$: unfrustrated (0-flux) for $J_{\pm} > 0$, frustrated (π -flux) for $J_{\pm} < 0$.

On a finite torus the periodic identifications generate additional short ice-preserving cycles that wrap the cluster. The dominant ones are four-site cycles threading four tetrahedra; they arise already at *second* order,

$$H_{\text{eff}}^{(\text{spur})} = -g_4 \sum_{\substack{\mathcal{C} \text{ 4-cycle} \\ \mathbf{w}_{\mathcal{C}} \neq \mathbf{0}}} (S_a^+ S_b^- S_c^+ S_d^-)_{\mathcal{C}} + \text{h.c.}, \quad g_4 = \frac{4J_{\pm}^2}{J_{zz}}, \quad (3)$$

with g_4 *sign-blind* in $J_{\pm}$ (second order) and

$$\frac{g_4}{g_{\text{hex}}} = \frac{J_{zz}}{3|J_{\pm}|} = \begin{cases} 3.3 & (J_{\pm} = -0.1 J_{zz}) \\ 6.7 & (J_{\pm} = +0.05 J_{zz}), \end{cases} \quad (4)$$

so the finite-size artifact is parametrically *larger* than the physics one wants to extract. Section 7 enumerates these cycles exactly: 36 on the $1 \times 1 \times 1$ cubic (16-site) cluster and 24 on the $2 \times 2 \times 2$ FCC (32-site) cluster, every one non-contractible ($\mathbf{w} \neq \mathbf{0}$); there are *no* contractible ice-preserving 4-cycles. Figure 1 contrasts a genuine bulk hexagon with a winding four-cycle.

Exactly at $J_{\pm} = -0.1 J_{zz}$, $\boldsymbol{\varphi} = \mathbf{0}$, the low spectrum of the 16-site cluster is (energies above the ground state, exact sparse ED in the $S_{\text{tot}}^z = 0$ sector)

$$E_n = \{0; 0.10603 (\times 4); 0.12595 (\times 9); \dots\}, \quad (5)$$

a *singlet* ground state with gap $\Delta = 0.10603 J_{zz}$ set by the g_4 scale, and the specific heat carries a spurious low- T Schottky peak at $T \simeq 0.037 J_{zz}$, on the four-loop scale $g_4 = 0.04 J_{zz}$ rather than the photon scale $g_{\text{hex}} = 0.012 J_{zz}$.¹

2 Twist phases attach to virtual paths, not to loops

2.1 Twist setup

The cluster is a flat 3-torus with first homology $\mathbb{Z}^3$. A $\text{U}(1)$ twist $\boldsymbol{\varphi} \in [0, 2\pi)^3$ is inserted, in the standard *boundary-bond (minimum-image) gauge*, by attaching a phase to every NN bond that

¹An earlier version of this manuscript quoted a “7-fold quasi-degenerate ground manifold” here, and degeneracies between 5 and 10 across twist corners. Exact re-diagonalization shows these were artifacts of degeneracy detection on Lanczos spectra: the correct ground-state degeneracies are 1 at $\boldsymbol{\varphi} = \mathbf{0}$ and at the three single- π corners, and 2 at the four remaining corners (table 2). The quoted *gaps* were correct.

Two emergent ring-exchange loops on the $1 \times 1 \times 1$ pyrochlore cluster

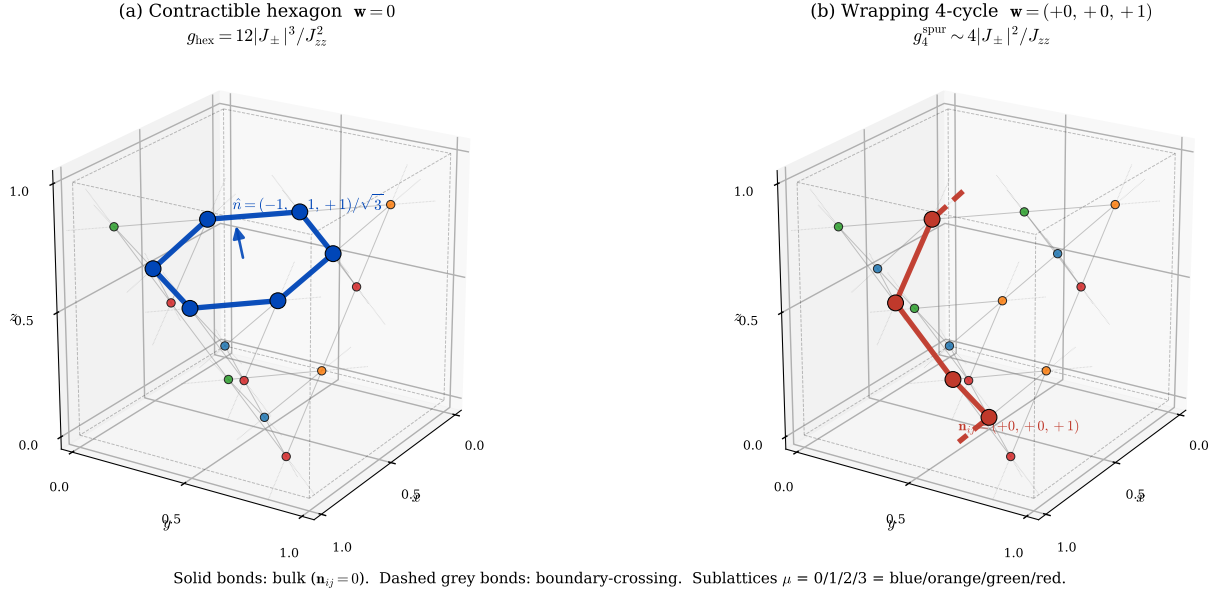


Figure 1: The two emergent ring-exchange loops on the $1 \times 1 \times 1$ pyrochlore cluster (16 sites, 8 tetrahedra, 48 NN bonds). **(a)** A genuine bulk hexagon ($\mathbf{w} = \mathbf{0}$) contributing the HFB ring exchange of eq. (2). **(b)** A four-site ice-preserving cycle with winding $\mathbf{w} = (0, 0, 1)$: absent on the infinite lattice, it generates the spurious second-order ring exchange eq. (3) on the torus.

crosses the periodic boundary, with integer wrap vector $\mathbf{n}_{ij} = \text{round}[(\mathbf{r}_j - \mathbf{r}_i)/L]$:

$$H_{\perp}(\varphi) = -J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- e^{-i\mathbf{n}_{ij} \cdot \varphi} + S_i^- S_j^+ e^{+i\mathbf{n}_{ij} \cdot \varphi}). \quad (6)$$

The Ising part is φ -independent, S_{tot}^z is conserved at every φ , and $\varphi = \mathbf{0}$ recovers eq. (1).

2.2 The matched-pair factorisation

It is tempting to argue – as an earlier version of this manuscript did – that every effective ring exchange over a closed loop $\mathcal{L}$ acquires the Wilson phase $e^{-i\mathbf{w}_{\mathcal{L}} \cdot \varphi}$ of its homology class, so that a twist average projects onto $\mathbf{w} = \mathbf{0}$ exactly. *This is false*, for an elementary reason: the phase of a perturbative process is the product of the phases of the *bond terms the process actually uses*, and the XXZ ring exchange never traverses the loop as a connected walk. The operator $O_{\mathcal{L}} = S_a^+ S_b^- S_c^+ S_d^-$ is a product of two commuting dimer moves, and there are exactly *two perfect matchings* of the cycle into NN dimer moves that generate it: $\{ab, cd\}$ (matching A) and $\{bc, da\}$ (matching B), each with two time orderings and equal energy denominators. Reinstating the twist phases per bond,

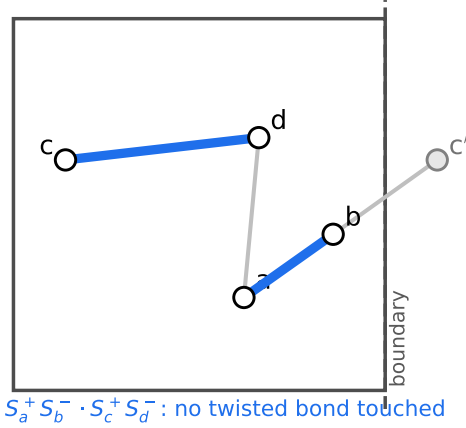
$$c_{\mathcal{L}}(\varphi) = -\frac{g_4}{2} \left[e^{-i\mathbf{N}_A \cdot \varphi} + e^{-i\mathbf{N}_B \cdot \varphi} \right], \quad \mathbf{N}_A = \mathbf{n}_{ab} + \mathbf{n}_{cd}, \quad \mathbf{N}_B = \mathbf{n}_{bc} + \mathbf{n}_{da}, \quad (7)$$

with

$$\mathbf{N}_A + \mathbf{N}_B = \mathbf{w}_{\mathcal{L}}. \quad (8)$$

$$c_C(\varphi) = -\frac{g_4}{2} [e^{-i\mathbf{N}_A \cdot \varphi} + e^{-i\mathbf{N}_B \cdot \varphi}] = -\frac{g_4}{2} (1 + e^{-i\mathbf{w} \cdot \varphi})$$

matching A: bulk bonds only, $\mathbf{N}_A = \mathbf{0}$



matching B: uses the crossing, $\mathbf{N}_B = \mathbf{w}$

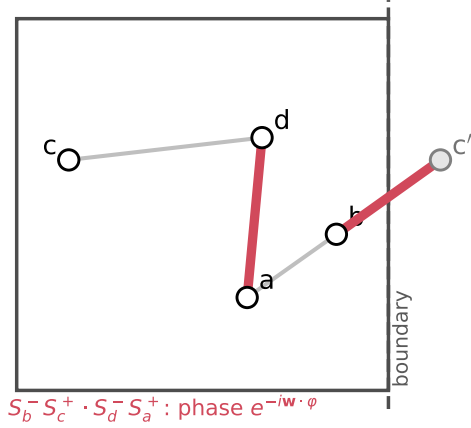


Figure 2: The two virtual paths (perfect matchings) that generate the ring exchange over a boundary-winding four-cycle $a-b-c-d$. Matching A (left) flips the dimers $\{ab\}$ and $\{cd\}$ using bulk bonds only: it carries no twist phase, $\mathbf{N}_A = \mathbf{0}$, and is invisible to *any* boundary twist. Matching B (right) uses the boundary-crossing bond and carries $\mathbf{N}_B = \mathbf{w}$. The coupling is the sum of the two.

Only the *sum* eq. (8) is fixed by homology; the individual matching windings $\mathbf{N}_{A,B}$ are properties of the virtual paths (and of the chosen gauge). The same structure holds for the hexagon at third order (three dimer moves, two matchings) and, suitably generalised, at every order. Figure 2 illustrates the two matchings of a winding four-loop.

Two exact corollaries follow.

(a) **Corner rule: couplings switch off, they do not flip sign.** The magnitude

$$|c_C(\varphi)| = \frac{g_4}{2} |1 + e^{-i\mathbf{w} \cdot \varphi}| = g_4 \left| \cos \frac{\mathbf{w} \cdot \varphi}{2} \right| \quad (9)$$

is gauge invariant. At a twist corner $\varphi = \pi \mathbf{p}$, $\mathbf{p} \in \{0, 1\}^3$, a winding loop's coupling is therefore $-g_4$ if $\mathbf{w} \cdot \mathbf{p}$ is even and *exactly zero* if $\mathbf{w} \cdot \mathbf{p}$ is odd. Corners do not build a $\pm$ interference pattern over loops; they *deactivate* subsets of loops. This is confirmed level-by-level by the exact corner spectra of table 2: the number of active four-loops falls from 36 (bare) to 20 ($|\mathbf{p}|_\pi = 1$) to 12 ($|\mathbf{p}|_\pi \geq 2$), and the low gap falls with it.

(b) **The bulk matching survives every twist average.** On both production clusters, *every* winding four-loop and every wrapping hexagon has one matching with $\mathbf{N} = \mathbf{0}$: its dimer moves carry zero net wrap (generically, both moves lie on bulk bonds, as in fig. 2), so its phase is φ -independent.² That matching is untouched by any boundary twist. Consequently the twist average

²Verified to machine precision by a path-resolved numerical Schrieffer–Wolff computation that tracks $\mathbf{N}$ for every virtual path: on the $1 \times 1 \times 1$ cubic and $2 \times 2 \times 2$ FCC clusters, exactly $1/2$ of the coefficient magnitude of every winding four-loop and every wrapping hexagon carries $\mathbf{N} = \mathbf{0}$, and *all* of the coefficient of every contractible hexagon carries $\mathbf{N} = \mathbf{0}$ (table 1).

Table 1: Path-resolved census of ring-exchange coefficients (numerical Schrieffer–Wolff, orders $J_{\pm}^2$ and $J_{\pm}^3$, exact coefficients). “8-corner op. avg.” is the surviving fraction of each coupling after averaging the effective *Hamiltonian* over $\{0, \pi\}^3$; “ $\delta = \mathbf{0}$ ” is the zero-transport projector of section 4. Every winding loop retains exactly half its coupling under any boundary-twist average; the transport projector is exact.

cluster	channel	#loops	8-corner op. avg.	continuum avg.	$\delta = \mathbf{0}$
(1, 1, 1) cubic, 16	winding 4-loops	36	1/2	1/2	0
	contractible hexagons	16	1	1	1
	wrapping hexagons	48	1/2	1/2	0
(2, 2, 2) FCC, 32	winding 4-loops	24	1/2	1/2	0
	contractible hexagons	32	1	1	1
	wrapping hexagons	96	1/2	1/2	0

Table 2: Exact corner anatomy of the 16-site cluster at $J_{\pm} = -0.1 J_{zz}$ (sparse ED, $S_{\text{tot}}^z = 0$). “Active 4-loops” is the number with $\mathbf{w} \cdot \mathbf{p}$ even [coupling $-g_4$; the others are switched *off* by eq. (9)]. The corner-mean gap is $0.0391 J_{zz} \approx g_4$: the artifact scale survives averaging.

corner class	corners	active 4-loops	gap Δ/J_{zz}	GS degeneracy
$ \mathbf{p} _{\pi} = 0$ (bare)	1	36	0.10603	1
$ \mathbf{p} _{\pi} = 1$	3	20	0.06404	1
$ \mathbf{p} _{\pi} = 2$	3	12	0.00372	2
$ \mathbf{p} _{\pi} = 3$	1	12	0.00372	2

of the effective *Hamiltonian* – over the eight corners, over any grid, or over the continuum – retains

$$\overline{H_{\text{eff}}^{(\text{spur})}} = -\frac{g_4}{2} \sum_{\mathbf{c}} (O_{\mathbf{c}} + O_{\mathbf{c}}^{\dagger}) + \dots, \quad (10)$$

half of every spurious coupling, at every order at which a bulk matching exists. Twist averaging in the boundary-bond gauge *cannot* remove the short-loop contamination, even in principle, even at the operator level. The good news is equally sharp: every contractible hexagon has *both* matchings on bulk bonds ($\mathbf{N}_A = \mathbf{N}_B = \mathbf{0}$), so the genuine ring exchange passes through all of this untouched, at full strength, at every twist.

2.3 Why the loop-holonomy argument fails: a self-correction

The superficially plausible identity $H_{\text{eff}}^{(n)}(\boldsymbol{\varphi}) = \sum_{\mathbf{w}} e^{i\mathbf{w} \cdot \boldsymbol{\varphi}} H_{\text{eff}, \mathbf{w}}^{(n)}(0)$ with $\mathbf{w}$ the *homology* winding of the flipped-site loop – the basis of the earlier version of this work, and of the intuition “averaging projects onto contractible loops” – would hold if the alternating $S^{\pm}$ factors around a loop always picked up the loop holonomy. They do not: the decomposition eq. (7) shows the operator receives contributions in *two* different twist-Fourier sectors $\mathbf{N}_A$ and $\mathbf{N}_B$, and only their sum is homological. The holonomy $e^{-i\mathbf{w} \cdot \boldsymbol{\varphi}}$ appears exactly once, as the *relative* phase of the two virtual paths – whence the gauge-invariant corner rule eq. (9) – but the *absolute* Fourier content is path-dependent, and the bulk matching sits at $\mathbf{N} = \mathbf{0}$ regardless of $\mathbf{w}$.

3 Averaging observables is weaker still

Even if one ignores the half-survival and averages, there is a second, independent error mode: in practice one averages *observables* [e.g. $\overline{C}(T) = \frac{1}{8} \sum_{\varphi} C(T, \varphi)$], not the Hamiltonian. The projection identities of section 2.2 are statements about objects *linear* in H_{eff} . The free energy is not linear, so its twist-Fourier components mix all winding sectors nonlinearly: corner-averaging an observable cancels only the terms *odd* in each spurious coupling, while all even powers survive at full strength.

A two-level caricature makes this sharp. Let the ice-manifold physics consist of one genuine scale $\varepsilon \sim g_{\text{hex}}$ and one spurious coupling that follows the corner rule, $g(\varphi) = g_4 \cos^2(\mathbf{w} \cdot \varphi/2)$: the gap is $\Delta(\varphi) = 2\sqrt{\varepsilon^2 + g(\varphi)^2}$. Averaging Δ (or any observable built from it) over corners leaves a mixture of $2\sqrt{\varepsilon^2 + g_4^2}$ and 2ε – the g_4^2 term *never cancels* – whereas the averaged *Hamiltonian* would carry $g = g_4/2$ (eq. (10)), and only the transport projector of section 4 reaches 2ε . Since $g_4/g_{\text{hex}} \gg 1$ [eq. (4)], the surviving even-power contamination *dominates* the photon scale.

The exact 16-site numbers land precisely on this hierarchy (fig. 3 and table 3): at $J_{\pm} = -0.1 J_{zz}$ the low- T peak sits at $0.0366 J_{zz}$ (bare), $0.0313 J_{zz}$ (corner-averaged observable: barely moves), $0.0178 J_{zz}$ (corner-averaged *operator*, i.e. the exact Löwdin-downfolded $H_{\text{eff}}(\varphi)$ averaged as a matrix over the eight corners: the half-coupling scale $g_4/2 = 0.02 J_{zz}$ of eq. (10)), and only the transport projector reaches $0.0075 J_{zz}$, at the photon scale. The corner-mean gap, $0.0391 J_{zz}$, is numerically indistinguishable from $g_4 = 0.04 J_{zz}$ – the observable average never left the artifact scale.

4 The zero-transport projector

4.1 The gauge-invariant selection rule

What labels a virtual process gauge-invariantly is not a winding class but its *net charge transport*

$$\delta = \sum_{\text{hops}} (\text{displacement of the moved boson}) = \underbrace{\sum_{i \text{ raised}} \mathbf{r}_i - \sum_{j \text{ lowered}} \mathbf{r}_j}_{\boldsymbol{\rho} \text{ (operator dipole)}} + \mathbf{N} \cdot \mathbf{L}, \quad (11)$$

the physical displacement vector of the S^z (boson) density moved by the process; δ is what a *uniform* vector potential couples to. In the smooth gauge – twist spread over every bond in proportion to its real displacement, $\theta_{ij} = \mathbf{A} \cdot \mathbf{d}_{ij}$ with $\mathbf{L}_k \cdot \mathbf{A} = \varphi_k$ – each process carries the phase $e^{-i\mathbf{A} \cdot \delta}$. Averaging the effective Hamiltonian over the *extended* twist period (the smooth-gauge Hamiltonian returns to itself, up to a diagonal gauge, only after $\varphi_k \rightarrow \varphi_k + 8\pi$ on the quarter-integer pyrochlore coordinates) is then a projector onto $\delta = \mathbf{0}$:

$$\boxed{\overline{H}_{\text{eff}}^{\delta} = \sum_{\text{processes: } \delta=\mathbf{0}} (\text{that process}).} \quad (12)$$

The selection rule eq. (12) does exactly what the naive homology projector was supposed to do, with a geometric argument that holds at *all orders*:

- **Winding four-loops die.** A four-loop matching transports two bosons by two NN dimer displacements, $|\delta_M| \leq 2d_{\text{NN}} = a/\sqrt{2} \approx 0.71a$, while $\delta_A + \delta_B = -\mathbf{w} \cdot \mathbf{L}$ with $|\mathbf{w} \cdot \mathbf{L}| \geq a$ (cubic) or $\sqrt{2}a$ (FCC $2 \times 2 \times 2$): neither matching can have $\delta = \mathbf{0}$. Higher-order paths generating the same operator can only shift δ by full cluster translations, and the operator dipole $\boldsymbol{\rho}$ is a quarter-integer vector too short to reach another translation: winding four-loops carry $\delta \neq \mathbf{0}$ in *every* realisation, to all orders.

- **Contractible hexagons survive at full strength.** The [111] pyrochlore hexagons are regular, so the alternating displacement sum of either matching vanishes identically: both virtual paths have $\delta = \mathbf{0}$ and pass the projector untouched.
- **Wrapping hexagons die** by the same length-count ($3d_{\text{NN}} \approx 1.06a$ cannot reach $|\mathbf{w} \cdot \mathbf{L}| \geq \sqrt{2}a$ on either cluster).

The census of table 1 (last column) confirms all three statements to machine precision at orders $J_{\pm}^2$ and $J_{\pm}^3$ on both production clusters: survival fractions exactly 0, 1, 0.

One honest bias remains at higher orders: a *contractible* loop process with a nonzero transport dipole ($\rho \neq \mathbf{0}$, possible for non-centrosymmetric loops at order $J_{\pm}^4$ and beyond) is also removed. Within the strict NN XXZ model the leading such objects are $O(J_{\pm}^4/J_{zz}^3)$ corrections, parametrically below g_{hex} ; we flag them rather than hide them.

4.2 Implementations

(i) Path-selected perturbation theory. A numerical Schrieffer–Wolff expansion in the ice manifold that tracks $(\mathbf{N}, \rho)$ per virtual path implements eq. (12) as a row filter; we carry it to fourth order including the $-\frac{1}{2}\{PVG^2VP, PVGVP\}$ correction. Cost: seconds (16 sites) to minutes (32 sites).

(ii) Exact downfolding plus dipole mask (recommended). Beyond perturbation theory the projector has a remarkably simple nonperturbative implementation. Diagonalize H *once* at $\varphi = \mathbf{0}$, Löwdin-project the lowest ice band onto the ice-configuration basis (the des Cloizeaux effective Hamiltonian; overlap singular values ≥ 0.58 on the 16-site cluster, so the band is cleanly ice-derived), and *zero out every matrix element whose flipped-site set carries a fractional dipole*: element (c, c') is kept iff $\rho = \sum_{i \text{ raised}} \mathbf{r}_i - \sum_{j \text{ lowered}} \mathbf{r}_j \equiv \mathbf{0} \pmod{\mathbf{L}}$. By eq. (11) this keeps exactly the $\delta = \mathbf{0}$ operator content, up to wrap-around ($|\mathbf{N}| \geq 1$) contributions to ρ -trivial elements, which are of higher perturbative order. No twist grid is needed at all. Validation against the path-selected construction (i): low- T peak 0.00122 vs 0.00127 J_{zz} at $J_{\pm} = -0.05$, and 0.00647 vs 0.00747 J_{zz} at -0.1 – where the mask value is the more reliable of the two, being nonperturbative (the difference is consistent with the $\sim 15\%$ fourth-order truncation of (i) at this coupling). On the 16-site cluster the mask retains 16% of the ice-basis matrix elements: the diagonal plus the contractible-hexagon connectivity.

(ii') Smooth-gauge averaging (conceptual form). The same projector is realised as an average of downfolded matrices over a smooth-gauge twist grid spanning the extended period $\varphi_k \rightarrow \varphi_k + 8\pi$; the corner-averaged-operator curve of fig. 3 is this construction on the $\{0, \pi\}^3$ subgrid and realises eq. (10) exactly. In practice the dipole mask (ii) supersedes it.

(iii) Counterterm cross-check. Since the winding loops are enumerable, one can add $+g_4 \sum_c (O_c + h.c.) + |\gamma_6| \sum_{\text{wrap hex}} (\dots)$ to the *bare* Hamiltonian and diagonalize exactly. This cancels the artifact only at leading order – precisely the objection raised against counterterms before – but it needs no downfolding and provides an independent exact-machinery reference; residual disagreement with (i)/(ii) measures the higher-order winding content.

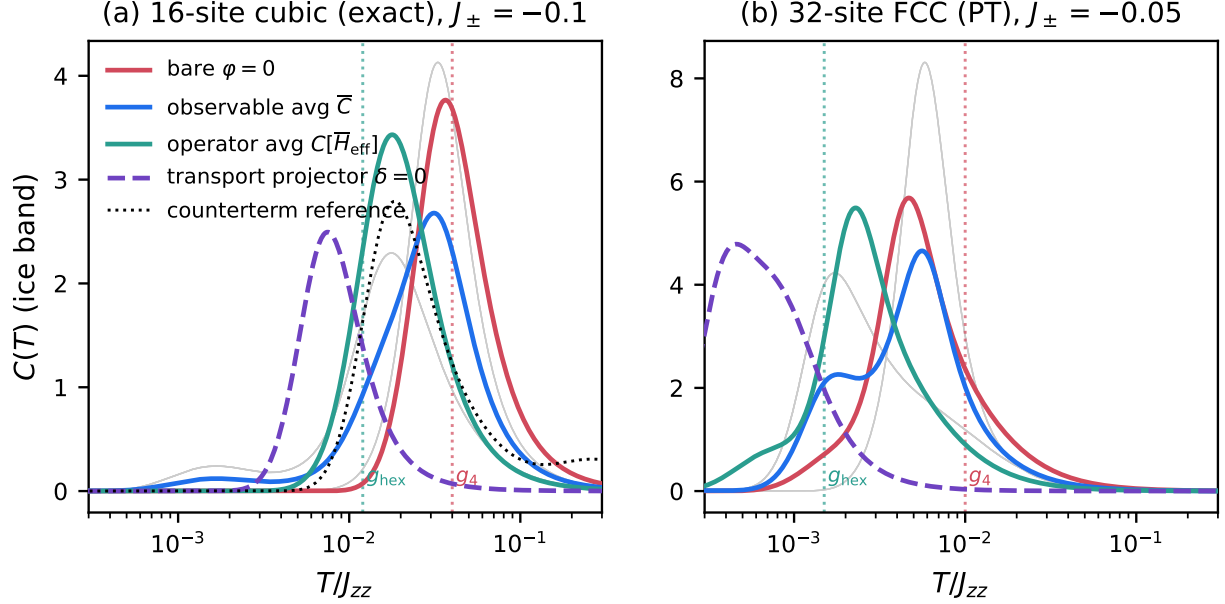


Figure 3: Ice-band specific heat under the four schemes. **(a)** 16-site cluster at $J_{\pm} = -0.1 J_{zz}$, everything exact: bare (red), corner-averaged observable $\bar{C}$ (blue; thin grey = individual corners), corner-averaged *operator* (teal; exact downfolded H_{eff} per corner, averaged as matrices), zero-transport projector (violet, dashed; path-selected H_{eff} through fourth order), and the counterterm reference (dotted). Vertical marks: g_4 and g_{hex} . The observable average barely moves the peak; the operator average stalls at the half-coupling scale $g_4/2$ [eq. (10)]; only the transport projector reaches the hexagon scale. (The projected curve is the path-selected H_{eff} through fourth order; benchmarked against exact spectra, the truncation error at $J_{\pm} = -0.1$ is $\sim 15\%$ of the low gap and shrinks as $J_{\pm}^5$; all other curves in (a) are exact.) **(b)** 32-site FCC cluster at $J_{\pm} = -0.05 J_{zz}$ (effective-Hamiltonian level, order ≤ 4): the corner-averaged observable splits its weight over corner gaps spanning two decades (four corners nearly gapless), leaving a low, smeared structure whose resolvable maximum still sits at the artifact scale while weight is pushed an order of magnitude below the photon peak; the transport projector lands at the hexagon scale $\approx 0.3 g_{\text{hex}}$.

5 Exact 16-site and 32-site demonstrations

We use $J_{xx} = J_{yy} = 0.2$, $J_{zz} = 1$ ($J_{\pm} = -0.1$) and, for the convergence checks, $J_{\pm} = -0.05$. All 16-site results are exact sparse ED in the $S_{\text{tot}}^z = 0$ sector (dimension 12870); the ice manifold holds 90 states. The 32-site FCC results use the path-resolved effective Hamiltonian (order $J_{\pm}^3$) in its 2970-state ice manifold.

5.1 16 sites: the hierarchy of schemes

Figure 3(a) and table 3 give the central result, already anticipated in section 3: observable averaging $0.037 \rightarrow 0.031 J_{zz}$ (a 15% move; the corner-mean gap remains at g_4), operator averaging $\rightarrow 0.018 J_{zz}$ ($= g_4/2$ scale, as eq. (10) demands), transport projector $\rightarrow 0.0075 J_{zz}$, below $g_{\text{hex}} = 0.012 J_{zz}$ and consistent with the finite-cluster hexagon-only model (the 16 contractible hexagons on 90 ice states). The counterterm reference lands at $0.019 J_{zz}$ at this coupling, and at $0.0025 J_{zz}$ versus the projector's $0.0013 J_{zz}$ at $J_{\pm} = -0.05$: the excess is the $O(J_{\pm}^4)$ winding content that a leading-order

Table 3: Low- T peak positions T_{peak}/J_{zz} of the ice-band $C(T)$ under each scheme. Reference scales: $g_4 = 4J_{\pm}^2$, $g_{\text{hex}} = 12|J_{\pm}|^3$. (16-site: exact ED; 32-site FCC: path-resolved H_{eff} ; δ -projector via path selection.)

cluster	$J_{\pm}/J_{zz}$	g_4	g_{hex}	bare	obs. avg.	op. avg.	δ -proj.	counterterm
16 cubic (exact)	-0.10	0.040	0.0120	0.0366	0.0313	0.0178	0.0075	0.0186
16 cubic (exact)	-0.05	0.010	0.0015	0.0130	0.0107	0.0063	0.0013	0.0025
32 FCC (PT ≤ 4)	-0.03	0.0036	0.00032	0.0017	0.0022	0.0009	0.0002	–
32 FCC (PT ≤ 4)	-0.05	0.010	0.0015	0.0047	0.0056	0.0023	0.0005	–
32 FCC (PT ≤ 4) [†]	-0.10	0.040	0.0120	0.0382	0.0231	0.0198	0.0029	–

[†]At $J_{\pm} = -0.1$ the frustrated-side SW series is not converged on the FCC cluster (the bare low gap moves by a factor ~ 8 from order 3 to order 4); this row is indicative only. At $J_{\pm} = -0.05$ the order-3 $\rightarrow$ 4 shifts are $\sim 10\%$ for the unprojected schemes and $\sim 40\%$ for the (much smaller) projected peak; the qualitative hierarchy is order-independent.

counterterm cannot cancel. Since that residue scales as $J_{\pm}^4$ while $g_{\text{hex}} \propto |J_{\pm}|^3$, its *relative* size grows as $J_{\pm} \rightarrow 0$ – counterterms are structurally unable to expose the photon scale, sharpening the objection raised against them in the earlier version of this work.

5.2 32 sites: why the corner-averaged low- T structure looks pathological

On the $2\times 2\times 2$ FCC cluster the corner rule deactivates winding loops corner-by-corner just as on 16 sites, but with only 24 loops (three times fewer per site) the surviving frustrated subsets produce wildly dispersed corner gaps: at $J_{\pm} = -0.05$ the eight corners have low gaps $\{5.2 \times 10^{-3}; 8.9 \times 10^{-5} (\times 4); 1.55 \times 10^{-2} (\times 3)\} J_{zz}$ – *more than two decades* of spread, with four corners nearly gapless. A corner-averaged $\overline{C}(T)$ is then a superposition of Schottky bumps at every one of these scales: its resolvable maximum stays at the artifact scale $[0.0056 J_{zz}$ vs the projected photon peak at $0.0005 J_{zz}$, table 3], its height collapses (the weight is split among incommensurate bumps), and a sizable fraction of the low- T weight is pushed an order of magnitude *below* the photon scale, to $T \sim 0.4 \times 9 \times 10^{-5} J_{zz}$ – beneath the temperature floor of typical FTLM grids ($T_{\text{min}} = 5 \times 10^{-3}$ in our production runs). In practice the “low peak” of a corner-averaged FCC run is therefore anomalously low and smeared, and none of its structure sits at the physical scale. The operator average again stalls at half coupling, and the transport projector lands at the hexagon scale, $0.0005 J_{zz} \approx 0.3 g_{\text{hex}}$, close to the bulk Schottky position $0.42 g_{\text{hex}}$ [fig. 3(b), table 3]. At larger couplings the same mechanism appears in a milder guise that matches production full-ED experience: bare and corner-averaged peak *positions* nearly coincide while the averaged peak is *lower* – the twist-invariant hexagons pin the position, and the height loss is the winding sector reacting to flux insertion (see section 9.1 for why this is itself a contamination measurement).

5.3 Entropy

The entropy benchmark of the earlier version survives unchanged in content: the exact number of ice states on the 16-site cluster is $N_{\text{ice}} = 90$ by direct enumeration, i.e. $S_{\text{ice}}/N = \ln(90)/16 = 0.2812$, 39% above the bulk Pauling estimate $\frac{1}{2} \ln \frac{3}{2} = 0.2027$, and the measured $S(T)$ shoulder at $T \sim 0.09\text{--}0.10 J_{zz}$ sits at 0.25–0.26 – the correct finite-size target, not the Pauling value (fig. 4). This feature lives above the ring-exchange scales and is insensitive to the averaging pathologies discussed here; we retain the FTLM curves.

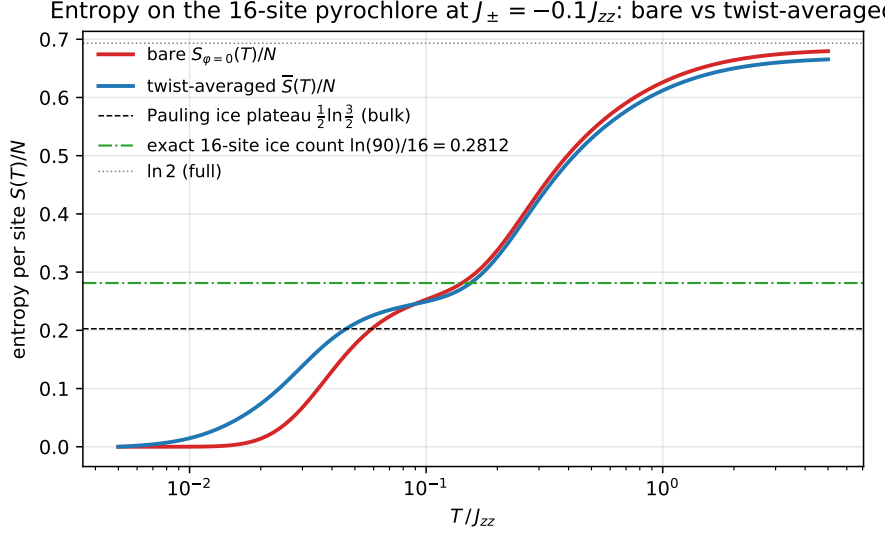


Figure 4: Entropy per site for bare and corner-averaged FTLM ED, with the bulk Pauling plateau 0.2027, the exact finite-cluster ice count $\ln(90)/16 = 0.2812$, and $\ln 2$ indicated. Both curves shoulder near the exact finite-size value. (FTLM stochastic uncertainty is a few percent; spectral quantities elsewhere in this paper are deterministic.)

6 The projected cluster as a laboratory: gauge-sector responses

The transport selection rule eq. (12) is not a property of the Hamiltonian alone: it classifies *every* operator acting in the ice sector. Any induced response operator – for instance the second-order action δH of a uniform transverse source $-\lambda \sum_i (S_i^+ + S_i^-)$, whose spinon-pair virtual processes are worked out in the companion source-response notes – decomposes into a $\delta = \mathbf{0}$ part (bulk gauge physics) and a winding part (finite-size artifact), and the *same* dipole mask projects it. This upgrades the projector from a spectral fix to a definition: *the bulk gauge sector of a small torus is the zero-net-transport sector of all its ice-manifold operators*, Hamiltonian and responses alike.

It also supplies the sharpest validation test we have. The measured source response of the bare cluster, $T_{\text{peak}}(\lambda) = T_{\text{peak}}(0)(1 - \kappa\lambda^2)$ with slope $B = \kappa|J_{\pm}|J_{zz}$, is known (companion notes) to be a near-cancellation of a bulk term (the flux-resolved spinon-vacuum polarizability, a rescaling of the gauge spectrum) against a winding-four-loop counter-term. If the projector removes exactly the winding sector, the projected cluster’s B must jump to the polarizability-only value. It does:

$J_{\pm} = -0.05 J_{zz}$	bare cluster	projected cluster	polarizability-only
B	+3.28	+9.55	+9.90

(the bare value matches the production full-ED measurement +3.20; the projected value is obtained by applying the dipole mask of implementation (ii) to *both* the downfolded Hamiltonian and the exactly computed response operator). Two conclusions. First, the projector removes the winding sector coherently across observables – spectrum and response together – to within a few percent. Second, and of independent physical interest, the projected cluster delivers a genuinely *bulk* gauge-sector quantity that the bare cluster mangles by a factor of three: the flux-resolved vacuum polarizability of the emergent U(1) (the response is 2–3 $\times$ stronger, with opposite sign, in

Table 4: Exact census of ice-preserving simple cycles (unchanged from the earlier version; independently re-verified).

cluster	N	tet.	bonds	wrap-bonds	4-cycles	contractible	hex (tot/c/w)
(1, 1, 1) cubic	16	8	48	18	36	0	64 / 16 / 48
(2, 2, 2) FCC	32	16	96	30	24	0	128 / 32 / 96

the 0-flux sector). Photon-scale thermodynamics was the projector’s first deliverable; bulk response functions are its second.

7 Combinatorics of the short loops

This section is unchanged in substance from the earlier version – the loop censuses were correct – and is retained for completeness, with the matching-resolved census of table 1 superseding the winding-only table as the operationally relevant one.

A simple cycle is ice-preserving iff every tetrahedron is touched by 0 or 2 of its sites. On the pyrochlore NN graph an ice-preserving 4-cycle is the boundary of a “square” of four distinct tetrahedra; such a square does not close in $\mathbb{R}^3$ (the girth of the ice-preserving loop set is six), so *every* ice-preserving 4-cycle wraps: 36 on (1, 1, 1) cubic [24 single-axis, e.g. $\mathbf{w} = (1, 0, 0)$, and 12 two-axis, e.g. $\mathbf{w} = (1, 1, 0)$] and 24 on (2, 2, 2) FCC [12:12]. The FCC cluster carries three times fewer spurious 4-cycles per site (0.75 vs 2.25) because its primitive translation $|\mathbf{a}_k| = a/\sqrt{2}$ spans two NN spacings. Hexagons: 64 ice-preserving [111]-planar 6-cycles on the cubic cluster, of which 16 are contractible (bulk density $\nu_c = 1$) and 48 wrap; 128 on FCC, 32 contractible ($\nu_c = 1$), 96 wrapping. The coupling of every 4-cycle is exactly $-4J_{\pm}^2/J_{zz}$ [two matchings $\times$ two orderings $\times$ denominator J_{zz} : a single NN hop creates two 3:1 tetrahedra at $J_{zz}/2$ each], and of every hexagon exactly $-12J_{\pm}^3/J_{zz}^2$, both reproduced to machine precision by the numerical Schrieffer–Wolff coefficients.

8 Cluster geometry: the bulk hexagon coupling is not renormalised

The geometric audit of the earlier version stands: on every faithful cluster the per-site density of contractible hexagons is the bulk value $\nu_c = 1$ (fig. 5), and the transport projector preserves exactly these hexagons at full strength (table 1). The residual finite-size error of the projected problem is therefore purely kinematic: the photon disperses linearly and the smallest nonzero cluster wavevector bounds the lowest photon energy, $\omega_{\min} \sim c k_{\min}$, so the projected Schottky peak sits above the bulk position $0.42 g_{\text{hex}}$ until $L N_{\varphi}$ is large. The 16-site projected peak at $0.0075 J_{zz} \approx 0.62 g_{\text{hex}}$ against the bulk $0.005 J_{zz}$ is consistent with this picture. Note that the “twist grids refine the Brillouin zone” bonus applies to the *observable* average in its free-particle sense and must not be conflated with artifact removal: the two goals now belong to two different constructions.

9 Discussion

Twist boundary conditions in QSI ED do less than advertised and more than expected, in a way this work makes precise.

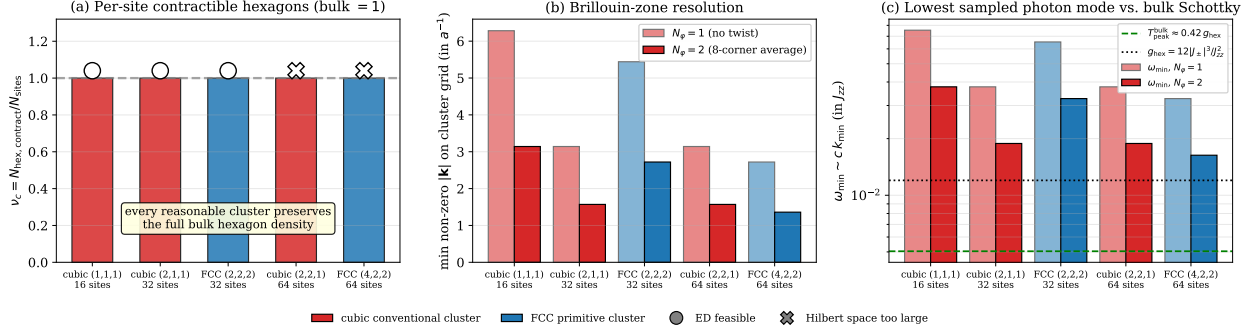


Figure 5: (a) Per-site contractible-hexagon density $\nu_c = 1$ on every faithful cluster. (b,c) Brillouin-zone resolution bound on the lowest accessible photon energy (unchanged from the earlier version).

Less than advertised: the classic picture – ring exchanges carry their homology holonomy, so twist averaging projects onto contractible loops – fails because ring exchanges are generated by *pairs of virtual paths*, and one path of every winding loop carries zero net wrap. Averaging the Hamiltonian over boundary twists removes at most half of each spurious coupling [eq. (10)]; averaging observables, as done in practice, removes only interference terms and leaves the artifact scale essentially intact (section 3). Both failure modes are confirmed exactly on the 16-site cluster, and they explain two long-standing puzzles of production runs: the twist-averaged 16-site peak that never approaches g_{hex} , and the anomalously low, smeared low- T structure of corner-averaged 32-site FCC runs (near-gapless frustrated corners push spectral weight far below the photon scale and below typical FTLM temperature floors).

More than expected: the corner rule eq. (9) says a single well-chosen twist switches off entire winding classes exactly [e.g. $\varphi = (\pi, \pi, \pi)$ kills all 24 single-axis four-loops on the 16-site cluster], and the corner-resolved spectra become a clean spectroscopy of the winding sector (table 2). The flux-dependence of that sector is physical: the four-loop channel is sign-blind in $J_{\pm}$ while the hexagon channel flips sign, so the two cooperate at 0-flux and compete at π -flux. This is directly measurable: in the companion transverse-source study, the measured source curvature of the gauge peak on the bare cluster contains a winding-four-loop component that counteracts the dominant spinon-continuum response; the four-loop component (of order ∓ 7 to ± 14 in the slope B , opposite on the two flux sides) is precisely the contamination that transport projection removes, turning the projected κ into a clean probe of the surviving spinon response.

The correct scheme is the zero-transport projector eq. (12): gauge-invariant, exact at all orders for the winding loops, transparent about its one bias (transport-dipole-carrying contractible processes at $O(J_{\pm}^4)$), and implementable either perturbatively (path selection) or nonperturbatively (exact downfolding plus the dipole mask). On the 16-site cluster it moves the low- T peak from the artifact scale $0.037 J_{zz}$ to the photon scale $0.0075 J_{zz}$ at $J_{\pm} = -0.1 J_{zz}$ – a factor of five – with the remaining distance to the bulk $0.42 g_{\text{hex}}$ accounted for by Brillouin-zone resolution, not by contamination.

9.1 When is the projection actually necessary? The 32-site PBC benchmark

An empirical challenge must be met head-on: full finite- T ED on the $2 \times 2 \times 2$ FCC cluster at $J_{\pm} \simeq -0.1 J_{zz}$, with plain periodic boundaries, already places the low- T peak in good agreement with the $g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2$ scale; corner-averaged and PBC results coincide in position there, the

averaged peak merely being lower (slower entropy release). If the largest tractable cluster gets the photon scale right without any machinery, what is the machinery for? Four answers, in increasing order of importance.

First, that benchmark sits at the most benign point of the whole parameter space, and the framework explains why. Three suppressions coincide at (FCC, π -flux, $J_{\pm} = -0.1$): the FCC cluster carries three times fewer winding loops per site than the cubic one (section 7); at π -flux the winding channel is additionally frustrated against the hexagons (table 1, companion analysis); and the contamination ratio $g_4/g_{\text{hex}} = J_{zz}/(3|J_{\pm}|)$ is *smallest* at the largest coupling. Agreement there confirms the census logic; it does not test the regime where the artifact lives.

Second, the necessity is a statement about the coupling range, and about scaling rather than one point. The artifact grows parametrically toward small $|J_{\pm}|$ – exactly where the QSI regime is robust and where one wants numerics. On the same FCC-32 cluster whose -0.1 point looks fine, the ratio of the bare peak to g_{hex} grows from ~ 1.5 (-0.1) to 3.1 (-0.05) to 5.4 (-0.03) [table 3], i.e. bare/projected reaches an order of magnitude, while at -0.03 four of the eight twist corners become essentially gapless ($4 \times 10^{-6} J_{zz}$). Equivalently: genuine photon thermodynamics must collapse as $|J_{\pm}|^3$; a g_4 -contaminated peak drifts toward $|J_{\pm}|^2$ scaling (the ratios above track $g_4/g_{\text{hex}} = J_{zz}/3|J_{\pm}|$), so agreement at a single coupling can be – and at -0.1 partly is – a crossing-point coincidence. The projector is what buys the *whole* coupling range and the scaling test.

Third, the observed twist-sensitivity is itself the measured contamination. The genuine gauge sector is provably twist-invariant – every contractible hexagon carries zero twist phase at every φ (section 2.2) – so if the -0.1 FCC physics were purely photonic, corner averaging would change *nothing*. The reported suppression of the averaged peak (“slower entropy release”) is the frustrated winding sector reacting to flux insertion: the corner rule switches loop subsets off, the corner gaps disperse (two decades at -0.05 , section 5), and the averaged entropy release smears. In other words, the very observation quoted against the machinery is a measurement of the residual artifact – twist response is the *diagnostic*, transport projection the *cure*.

Fourth, peak position is the least sensitive observable. Even where the bare peak position looks right, gauge-sector *responses* remain wrong by $O(1)$ factors: the source-response slope of the bare cluster is a near-cancellation ($B = +3.3$ vs the bulk-relevant $+9.9$ at $J_{\pm} = -0.05$; section 6), i.e. off by a factor of three at couplings where $C(T)$ looks converged. And beyond 32 sites full finite- T ED ends, while the projector’s perturbative implementation (path selection in the ice manifold) scales to the 48- and 64-site clusters whose Brillouin zones begin to resolve the photon (fig. 5) – PBC luck at 32 sites offers no such path.

9.2 Objective: a benchmarked laboratory for the π -flux gauge sector

The projector’s ultimate purpose is not a cleaner specific-heat peak but access to physics that is otherwise out of reach. Three facts combine. (i) The effective ring-exchange (gauge) Hamiltonian lives in the 2970-state ice manifold of the $2 \times 2 \times 2$ FCC cluster, so its *full* spectrum – and hence numerically exact gauge-sector thermodynamics, $C(T)$ and $S(T)$ down to $T \rightarrow 0$ – is obtained by dense diagonalization, without the low- T ceiling of stochastic finite- T methods. (ii) The transport projector removes the winding contamination that otherwise corrupts this spectrum, and does so for *every* ice-manifold operator, not only H (section 6). (iii) The QSI transverse coupling is sign-problem-free for QMC only at $J_{\pm} > 0$; the π -flux sector ($J_{\pm} < 0$) of the cerium pyrochlores is a QMC dead zone.

This defines a concrete program, and the natural culmination of the present method. Benchmark the projected 0-flux gauge thermodynamics against published sign-problem-free QMC at the same coupling; then cross into π -flux, where the target observable is the flux-locked response of the emer-

gent photon to a uniform conjugate field – a background-free thermodynamic discriminant of the gauge flux (softening at π -flux, hardening at 0-flux), computable from the same transport-projected ice-manifold operators and measurable by dilatometry or magnetocalorimetry. The couplings scale as $J_{\pm}^k$ exactly, so a single effective-Hamiltonian build serves the entire sweep and both flux signs; the campaign, resource estimates, and go/no-go gates are detailed in the accompanying simulation plan.

Reproducibility

All scripts live in `twist.qsi_demo/scripts/`. The path-resolved Schrieffer–Wolff engine (orders 2–4, winding and transport tracking, both clusters) is `ice_pt_lib.py`; exact sparse ED, ring-operator sums, Löwdin downfolding and the dipole-mask projector (`SzBasis.transport_mask`) are in `exact_ed_lib.py`; the exact source-response machinery behind section 6 is `kappa_lambda2_exact.py`; `pt_validate.py` validates the effective Hamiltonians against exact spectra (and documents the degeneracy corrections of eq. (5)); the matching census of table 1 is `winding_census.py`; the scheme comparison of fig. 3 and table 3 is `operator_vs_observable.py`; the channel-weight analysis cited in the discussion is `kappa_flux_analysis.py` (figures in `gauge_probe_prl/notes/`). Legacy FTLM drivers (`run_demo.py`, `analyze_demo.py`) reproduce the entropy figure; their spectral post-processing should not be used for degeneracy counting (see section 1).

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